

NARAMREDDY CHARAN KUMAR REDDY

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PROFESSIONAL SUMMARY

3rd-year Computer Science student specializing in AI and Machine Learning, building end-to-end systems that reach production rather than stopping at a notebook metric. Experience spans medical-imaging deep learning, gradient-boosted forecasting across thousands of parallel series, and recommendation architectures on learned embeddings — plus the serving layer around them: FastAPI, Docker and cloud deployment. Strong bias toward leakage-free validation and metrics that describe real behaviour.

TECHNICAL SKILLS

Machine Learning:	XGBoost, LightGBM, Random Forest, Logistic Regression, Scikit-Learn, Optuna.
Deep Learning & CV:	PyTorch, Transfer Learning (ResNet / DenseNet / EfficientNet), CNNs, pydicom, Neural Collaborative Filtering, Vector Embeddings, FAISS.
Data Intelligence:	Pandas, NumPy, Feature Engineering, KNNImputer, Imbalance Handling, RobustScaler.
Engineering & Deploy:	Python 3 (Advanced OOP), Data Structures, FastAPI, REST APIs, SQL (Schema Design), SQLAlchemy, Docker, Azure Container Apps, Git.

ENGINEERING PROJECTS

RSNA Pneumonia Detection | Chest X-Ray Screening | PyTorch, pydicom, FastAPI, [NaramCharan/RSNA-Pneumonia-Detection](https://github.com/NaramCharan/RSNA-Pneumonia-Detection)
React, Docker, Azure

Live application: rsna-app.salmonmeadow-7644e67e.eastasia.azurecontainerapps.io

- Benchmarked DenseNet-121, EfficientNet-B2 and ResNet-34 under an identical two-stage transfer-learning protocol; the fine-tuned ResNet-34 reached **83% pneumonia recall, 0.79 F1 and 86% accuracy** on 1,836 held-out studies, against a 68% always-negative baseline.
- Diagnosed a RandomCrop(224) transform discarding 95% of every 1024×1024 radiograph while retaining the study-level label — removing it produced a larger gain than every architecture change in the project combined.
- Prevented patient-level leakage by splitting on unique patient IDs rather than rows and evaluated the test set exactly once; handled 70/30 class imbalance with weighted CrossEntropyLoss and F1-driven scheduling and early stopping.
- Deployed end-to-end: one Docker container on Azure Container Apps serving a React interface and the REST API from a single origin; removed an unnecessary PostgreSQL layer to cut hosting cost 84% (\$30.74 → \$5.06/month).

Walmart Store Weekly Sales Forecasting | Python, Scikit-Learn, XGBoost, [NaramCharan/Walmart-Store-Weekly-Sales-Forecasting](https://github.com/NaramCharan/Walmart-Store-Weekly-Sales-Forecasting)
LightGBM, Pandas

- Engineered a global multi-series forecasting pipeline predicting weekly sales across ~3,000 store-department combinations, achieving **95.55% validation R²** with an optimized LightGBM regressor.
- Built a custom recursive walk-forward inference system from scratch to project rolling and lag features over future validation horizons where actual sales are missing.
- Eliminated critical temporal data leakage by splitting before feature extraction and computing lag/rolling statistics strictly within each partition, ensuring valid validation.

E-Commerce Customer Churn Prediction | Python, Scikit-Learn, XGBoost, [NaramCharan/ecommerce-customer-churn-prediction](https://github.com/NaramCharan/ecommerce-customer-churn-prediction)
PyTorch, Optuna

- Engineered an end-to-end classification pipeline using ColumnTransformer with hybrid KNN/simple imputation to clean, scale and encode customer data while strictly preventing data leakage.
- Tuned tree classifiers with Optuna over 10-fold Stratified Cross-Validation, achieving **98.28% accuracy and 94.74% F1** on the final selected XGBoost production model; a 4-hidden-layer PyTorch network scored 98.76% and was rejected as not worth the compute.

EDUCATION

B.Tech in Computer Science Engineering (AI & Machine Learning) — 3rd Year
GD Goenka University — Gurugram, India

Graduation: May 2028
Current CGPA: 8.98 / 10.0

CERTIFICATIONS

- Deep Learning Specialization – DeepLearning.AI (July 2026)
- Claude Code: A Highly Agentic Coding Assistant – DeepLearning.AI, Anthropic (July 2026)
- Machine Learning Specialization – DeepLearning.AI, Stanford University (May 2026)
- Databases and SQL for Data Science with Python – IBM, Coursera (Nov 2025)
- Python for Everybody Specialization – University of Michigan, Coursera (March 2025)
- Prompt Engineering & Generative AI – Google, Vanderbilt University (March 2025)